

STAT517: Statistical Inference

Course Information

Course number and title: STAT 51700 Statistical Inference

Meeting time: Online, Asynchronous

Course credit hours: 3 credits

Prerequisites: STAT 516 or STAT 519. In addition, an understanding of undergraduate-level calculus is required.

Instructor Contact Information

Instructor: Jin Hyung Lee

Email: lee6024@purdue.edu

Faculty Website: <https://jlee309.github.io/>

Office Hours: Instructor's online office hours are held at 10:00 AM to 11:00 AM every Tuesday.

Zoom Link :

<https://purdue-edu.zoom.us/j/97727063272?pwd=6d0CAfTI7xnqV0xHb3AJRr538r5zqa.1>

Time Zone: Be aware that all times stated in this class are for the Eastern time zone (W. Lafayette local time).

TA Contact Information

TA: Arin Chang

Email : chan1074@purdue.edu

Office Hours: TA's online office hours are held at 5:00 PM to 6:00 PM every Wednesday and Thursday.

Zoom link: <https://purdue-edu.zoom.us/j/9620676171>

For grading-related inquiries, please contact TA as your first point of contact.

Any grading inquiries or disputes must be discussed within one week of the grade being posted. Assignments are typically graded by the following Wednesday. Please note that disputes regarding earlier assignments will not be accepted at the end of the semester.

Instructor's Online Hours

I will be available and respond to student questions as soon as **I am available (generally within 48 hours) during the M–F work week.** Student inquiries made during the weekend may experience a delayed response time. Questions about the course content, assignments, or lectures should be asked in the **Course Q & A** forum provided on the discussion boards. Students are encouraged to answer the questions their peers ask on the **Course Q & A** forum. Email should

only be used for personal questions. **When emailing me, please place the course number in the subject line of the email.** This will help me tremendously in locating your emails quicker.

Academic Dishonesty

Purdue prohibits “dishonesty in connection with any University activity. **Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty.**” [Part 5, Section III-B-2-a, University Regulations] Furthermore, the University Senate has stipulated that “the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest.” [University Senate Document 72-18, December 15, 1972]

Use of AI / Large Language Models

You are permitted to use generative AI tools and large language models (LLMs), such as ChatGPT or Claude, as a study aid while learning the course material. For example, you may use them to clarify concepts, explore alternative explanations, or check your understanding of the lecture content. **However, the use of any AI or LLM tools is strictly prohibited on all graded work, including homework assignments, online discussions, the midterm, and the final exam. All work submitted for these components must be entirely your own. Using AI tools on graded work will be treated as a violation of the Academic Dishonesty policy described above.**

Course Description

This is a master level mathematical statistical course covering standard statistical methods and their applications. Topics include unbiased estimation, maximum likelihood and moment estimation, testing hypotheses for standard distributions and contingency tables, confidence intervals and hypothesis testing, introduction to concepts of sufficiency, loss functions, and risk functions, and Bayesian statistics.

This is an 8-week fully online course. Each week includes 7–10 lectures and 1 homework. The online discussion problems are given approximately every week.

The course PDF materials were generated using Latex and may not be fully accessible. If students have a documented accommodation, contact the instructor immediately to work with them to get them an alternative format.

Course Outcomes

- CO1: Review probability
- CO2: Employ some elementary statistical inferences such as confidence interval, and hypothesis testing
- CO3: Understand the basic properties of estimators such as consistency, limiting distributions, and the central limit theorem

- CO4: Master how to apply the maximum likelihood inference to solve statistical problems
- CO5: Recognize the concepts of sufficiency, loss, risk, completeness, minimum variance unbiased estimators
- CO6: Perform the Bayesian inference, including the prior, statistical model, the posterior, credible intervals, and hypothesis testing.

Technical Requirements

The following information has been provided to assist you in preparing to use technology successfully.

- A reliable internet connection – capable of consistently streaming video and stable enough to finish short exams without dropping connection.
- Access to Purdue's Brightspace Learning Management System – All course content, course readings, and exams will be accessed online through Brightspace.
- For homework and exam submission, you must take a picture of your work and upload it as a PDF file to Brightspace. A phone camera is required to take the photo.

Learning Resources & Texts

Required Textbook:

1. *Introduction to Mathematical Statistics* by Hogg, McKean, Craig, 8th edition (ISBN-13: 978-0134686998; ISBN-10: 0134686993)

Optional Textbook:

2. *Probability and Statistics: The Science of Uncertainty*, Second Edition (ISBN-13: 978-1429224628; ISBN-10: 1429224622)
3. *John E. Freund's Mathematical Statistics with Applications*, 8th Edition (ISBN-13: 978-0321807090 ISBN-10: 032180709X)

Assignments

You will have 7 graded assignments and 7 graded online discussions throughout the semester. Details on these assignments, and guidelines on discussion participation and evaluation will be posted on Brightspace. The due dates for the assignments posted on Brightspace are in Eastern Standard Time (the local time zone of West Lafayette, Indiana).

Assignments	Points
Assignments	50 %
Graded Discussions	10%
Midterm	15%
Final Exam	25 %
Total	100%

Students are required to submit their homework solutions through Brightspace before the due date. The grader will grade the homework and return them to students within one week of submission.

Students are expected to actively participate in online discussions within one week of each post. Both the instructor and the TA will review the discussion answers and provide constructive feedback for each response. Your active participation counts 10% toward your final grade. Providing solutions to the online discussion, raising and answering questions are examples of active participation.

There are two open-book online exams that you will have a 48-hour window to work on them. However, once you start the exam, you only have 120 minutes to finish each exam.

Attendance Policy

Students are expected to watch the lecture videos, complete weekly homework on time. Late submission will not be accepted. Please check the requirements for each of the assignment on the course site in Brightspace.

This course follows the Academic Regulations: Attendance and Office of the Dean of Students: Class Absences posted in Brightspace under “University Policies and Statements.” The policies state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. **When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification is not possible, contact me as soon as possible by email.**

For cases that fall under the University excused absence regulations – Grief/Bereavement, Military Service, Jury Duty, Parenting Leave, or the Medically Excused Absence Policy for Students – you or your representative go to the Office of the Dean of Students (ODOS) website to complete appropriate request forms. ODOS reviews these requests and, if granted, will notify all your instructors. In cases related to COVID-19, please follow the Protect Purdue Updates for the Spring 2023 Semester.

Course Schedule

The course schedule is subject to change as the semester progresses. Please check Brightspace regularly for updates.

Week	Modules	Lectures	Assignments
Week 1	Module 1-1	Lecture 1: Classical Probability Lecture 2: Random variables Lecture 3: Transformation and expectations	• Assignment 1 • Welcome and Introductions • Online Discussions 1
Week 1	Module 1-2	Lecture 1: Multivariate random variables Lecture 2: Transformation for multivariable RVs Lecture 3: Conditional distributions Lecture 4: Dependence and correlation	

Week	Modules	Lectures	Assignments
Week 2	Module 2-1	Lecture 1: Sampling and Statistics Lecture 2: t and F distributions, and student's theorem Lecture 3: Confidence intervals Lecture 4: Problem Solving	• Assignment 2 • Online Discussions 2
Week 2	Module 2-2	Lecture 1: Introduction to hypothesis testing Lecture 2: Hypothesis testing procedures Lecture 3: P-value Lecture 4: Problem Solving	
Week 3	Module 3-1	Lecture 1: Order Statistics and their Distributions Lecture 2: Important Statistics related to Order Statistics Lecture 3: Order Statistics for Discrete Random Variables Lecture 4: Problem Solving	• Assignment 3 • Online Discussions 3
Week 3	Module 3-2	Lecture 1: Chi-Square Tests Lecture 2: The Method of Monte Carlo Lecture 3: Problem Solving	
Week 4	Module 4-1	Lecture 1: Definition of convergence in probability Lecture 2: Consistency Lecture 3: Problem Solving	• Assignment 4 • Online Discussions 4
Week 4	Module 4-2	Lecture 1: Definition of convergence in distribution Lecture 2: The Delta-method and the mgf technique Lecture 3: The central limit theorem Lecture 4: Problem Solving	
Week 5	Module 5-1	Lecture 1: Practice Midterm	• Practice Midterm • Midterm (July 16-17)
Week 5	Module 5-2	Lecture 1: The concepts of MLEs Lecture 2: The computing of MLEs Lecture 3: The properties of MLEs Lecture 4: Problem Solving	• Assignment 5
Week 6	Module 6-1	Lecture 1: Fisher information Lecture 2: Rao-Cramer lower bound Lecture 3: Asymptotic normality Lecture 4: Problem solving	

Week	Modules	Lectures	Assignments
Week 6	Module 6-2	Lecture 1: Likelihood ratio tests Lecture 2: Other related tests Lecture 3: Multiparameter case: estimation Lecture 4: Multiparameter case: testing Lecture 5: Problem solving	• Assignment 6 • Online Discussions 5
Week 7	Module 7-1	Lecture 1: Decision theory Lecture 2: Definition of sufficient statistics Lecture 3: The factorization theorem Lecture 4: Properties of sufficient statistics Lecture 5: Problem solving	• Assignment 7 • Online Discussions 6
Week 7	Module 7-2	Lecture 1: The definition of a complete family Lecture 2: Completeness and MVUE Lecture 3: Find MVUE Lecture 4: The exponential class	
Week 8	Module 8-1	Lecture 1: Philosophy Lecture 2: Decision Theory Lecture 3: An Example Lecture 4: Computation	• Assignment 8 (optional) • Online Discussions 7 (optional) • Practice Final Exam • Final Exam (August 6-7)
Week 8	Module 8-2	Lecture 1: Practice Final Exam	

Grading Scales

The Grading Scale in this course follows the Typical A–F structure, with +/– as appropriate. For full detail, this has been provided below:

<i>Percentage</i>	<i>Letter Grade</i>
95% or higher	A+
90% – 94%	A
85% – 89%	A-
80% – 84%	B+
75% – 79%	B
70% – 74%	B-
65% – 69%	C
60% – 64%	D
59.9% or lower	F

Netiquette

You are encouraged to comment, question, or critique ideas. **However, be mindful that sarcasm and humor can be easily misconstrued in online interactions.** Please read the Netiquette rules for this course:

- Give other students the opportunity to join in the discussion.
- Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Keep an “open-mind” and be willing to express even your minority opinion.
- Think and edit before you share (e.g., post or email).
- Ask for feedback.

Course Evaluation

During the last two weeks of the course, you will be provided with an opportunity to evaluate this course and your instructor. Purdue now uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to two weeks to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Emergency Statement

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor’s control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email. You are expected to read your @purdue.edu email on a frequent basis.

Disability Statement

Students with disabilities must be registered with Disability Resource Center in the Office of the Dean of Students before classroom accommodations can be provided. If you are eligible for academic accommodations because you have a documented disability that will impact your work in this class, please schedule an appointment with me as soon as possible to discuss your needs.

Nondiscrimination

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

Academic Guidance in the Event a Student is Quarantined/Isolated

If you become quarantined or isolated at any point in time during the semester you will have access to an Academic Case Manager who can provide you academic support. Your Academic Case Manager can be reached at acmq@purdue.edu and will provide you with general guidelines/resources around communicating with your instructors, be available for academic support, and offer suggestions for how to be successful when learning remotely. Importantly, if you find yourself too sick to progress in the course, notify your academic case manager and notify me via email or Brightspace. We will make arrangements based on your particular situation. The Office of the Dean of Students (odos@purdue.edu) is also available to support you should this situation occur.